MOON MAGIC!

Phases, Eclipses and Hands-On Fun





A kid-friendly guide to why the Moon changes shape and why eclipses happen

Meet the Space Team



The Sun

- A **star**, a giant ball of super-hot glowing gas.
- Makes its **own** light and heat.
- So big that about 1.3 million Earths would fit inside!



The Earth

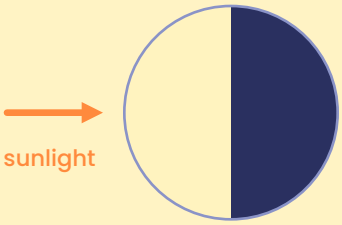
- Our home planet.
- Spins once every **24 hours**, which gives us day and night.
- Travels around the Sun once a year.



The Moon

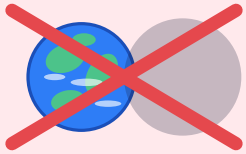
- Earth's space neighbor, about 384,000 km away.
- Makes **no light** of its own. It bounces sunlight, like a mirror.
- Goes around Earth about once a month.

BIG IDEA



The Moon is like a giant night-light that **bounces sunlight**. **Half of the Moon is always lit by the Sun**, just like half of Earth is always having daytime. A **phase** is simply **how much of that lit half we can see** from Earth.

MYTH BUSTER

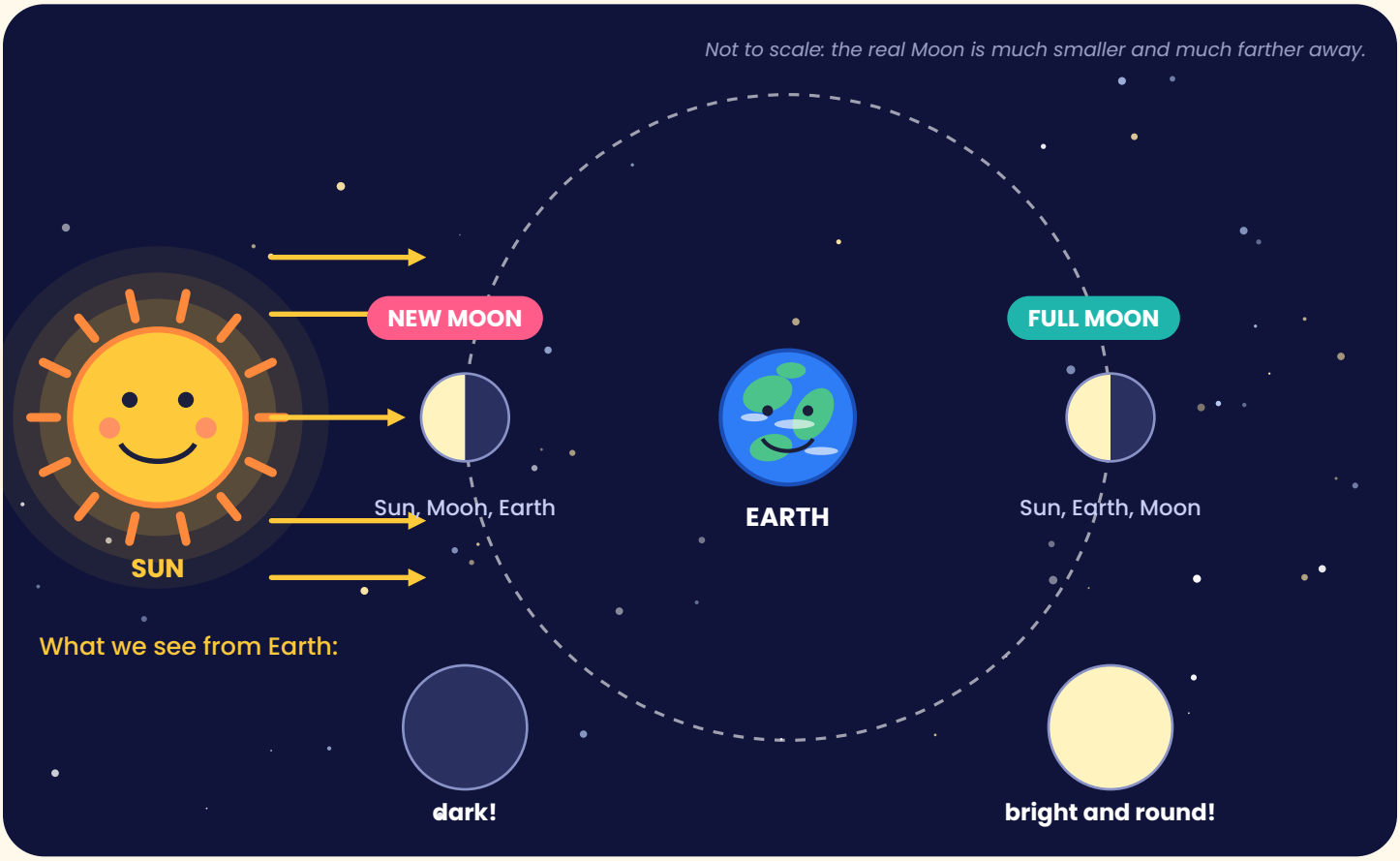


Myth: "Moon phases happen because Earth's shadow covers part of the Moon."

Truth: Nope! Earth's shadow only touches the Moon during a **lunar eclipse** (page 8). Phases happen because we see more or less of the Moon's **sunlit half** as it travels around Earth.

New Moon and Full Moon

Imagine floating high above space and looking **down** on the Sun, Earth and Moon. The Moon walks around Earth in a big circle. Here are its two most special spots:



New Moon

The Moon is **between** the Sun and Earth. The Sun lights up the half of the Moon that faces *away* from us. The half facing Earth is in shadow, so we see a dark sky where the Moon should be.

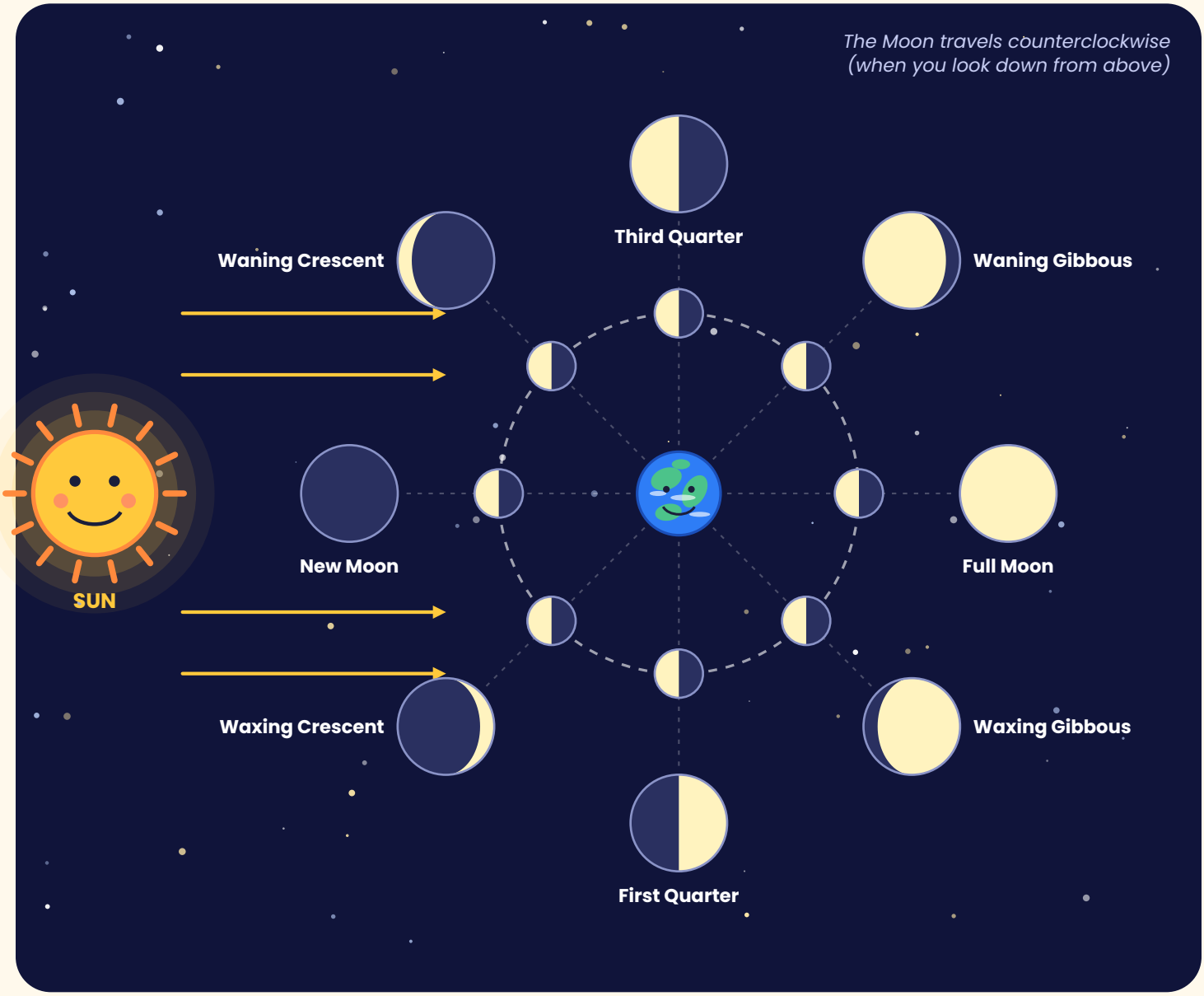
Full Moon

Earth is **between** the Sun and the Moon (the Moon passes a little above or below Earth's shadow). The Sun lights up the half facing Earth, so we see the **whole bright circle**!

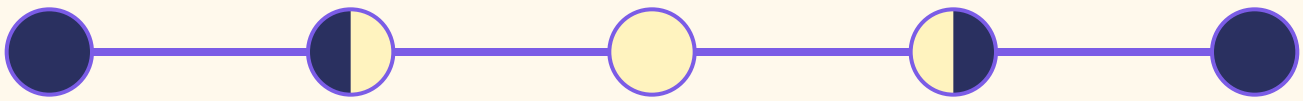
Remember it! S – M – E (Sun, Moon, Earth) is a **New** Moon. **S – E – M** (Sun, Earth, Moon) is a **Full** Moon.

All 8 Moon Phases

The Moon takes about **29 and a half days** to go from one New Moon to the next. The **inner ring** shows the Moon in space. The **outer ring** shows what you would see from Earth!



A month in the life of the Moon



Day 0
New

Day 7
First Quarter

Day 15
Full

Day 22
Third Quarter

Day 29.5
New again

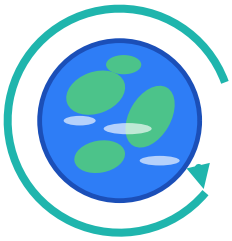
Waxing means growing (the lit part gets bigger). **Waning** means shrinking. **Gibbous** means more than half lit.

24 Hours or 27 Days?

THE BIG QUESTION

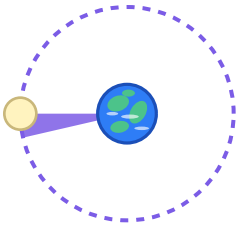
Can the Moon be **between** the Sun and Earth, and then **on the far side** of Earth, all within 24 hours?
No! 24 hours is how long **Earth** takes to spin once (one day and night). The **Moon** takes about **27 days** to travel once around Earth.

Earth spinning (24 hours)



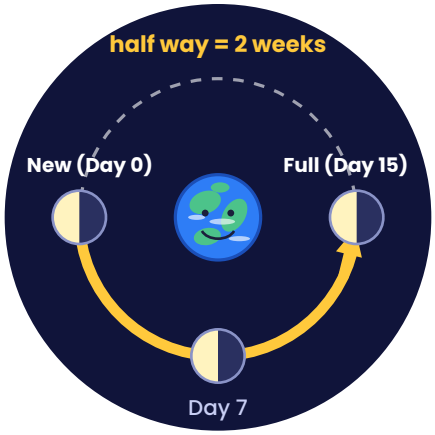
Earth does a **full 360 degree** spin.
That is **1 day** and **1 night**.

Moon traveling (24 hours)



In 24 hours the Moon only moves about **13 degrees** along its path. That is just a thin slice!

How far to the other side?



Kid math time

- The Moon zooms about **1 km every second**.
- One day is **86,400 seconds**, so it travels about **86,400 km a day**.
- Half way around is about **1,200,000 km**.
- 1,200,000 divided by 86,400 is about **14 days**.

To do it in **1 day**, the Moon would have to go **14 times faster**!

BONUS QUESTION

Why does the Moon rise later every night?

Every day the Moon moves about 13 degrees further along its path. Earth has to spin a little extra, about **50 minutes**, to catch up. So tonight the Moon rises about **50 minutes later** than it did last night!



Demo 1: Lamp and Ball

What you need

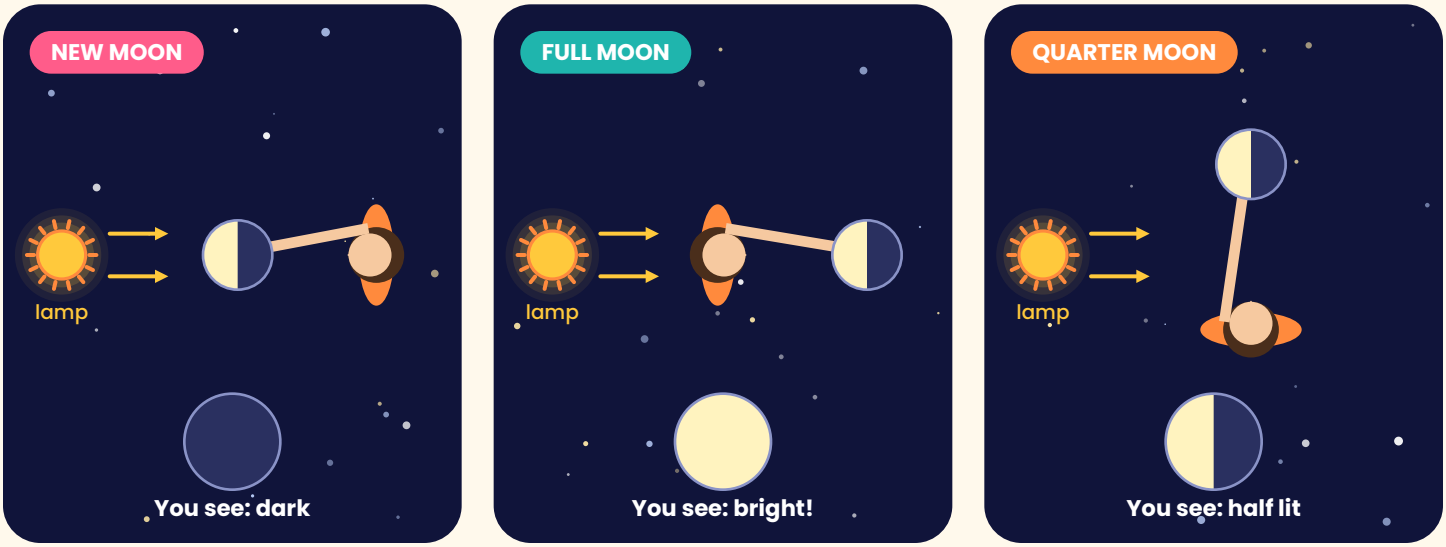
Lamp or flashlight
is the SUN

A light-colored ball
is the MOON

You!
are EARTH

A dark room
so the light shows

Looking down from above



Try it!

1. Turn off the other lights. Put the lamp on a table. It is the **Sun**.
2. Stand about two big steps away, facing the lamp. Your head is **Earth**.
3. Hold the ball out at arm's length, a little **higher than your head**. It is the **Moon**.
4. **New Moon:** face the lamp, so the ball is between the Sun and you. You see its dark side!
5. Now turn slowly to your **left**, keeping the ball in front of you. Watch a crescent, then half, then more. When your back is to the lamp, the whole ball glows: **Full Moon!** Keep turning to see all 8 phases.

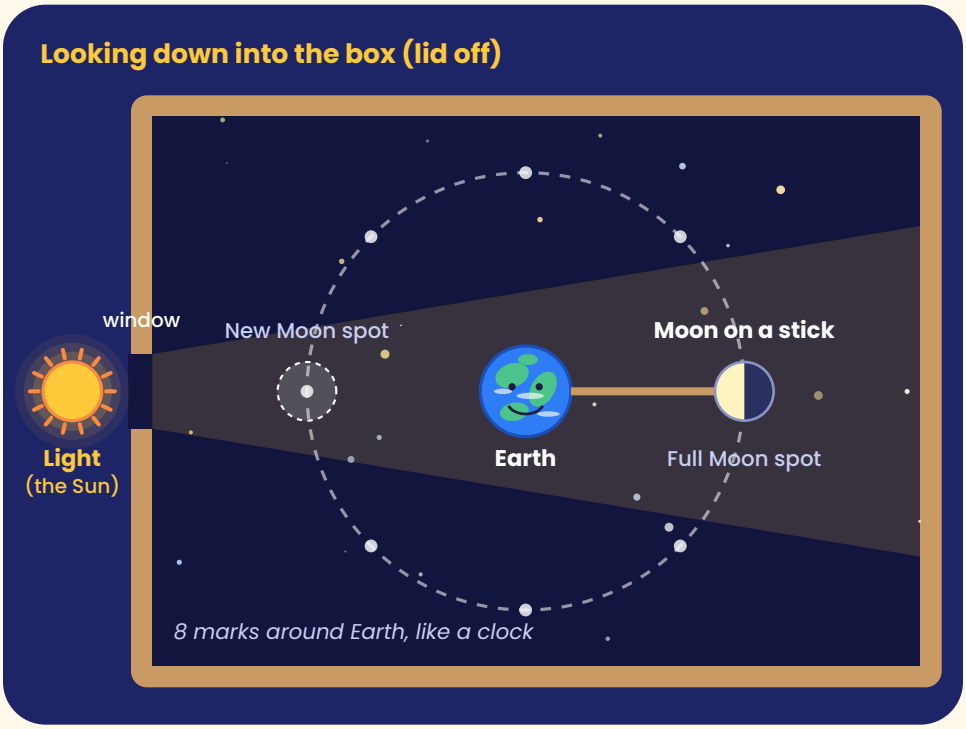
Why it works

The lamp always lights the half of the ball that faces it. As you turn, you see more or less of that lit half, exactly what happens to us on Earth!

Tips

- Use an LED lamp or flashlight. Hot bulbs can burn.
- Hold the ball high so your own shadow does not fall on it. (That would be an eclipse!)
- Take turns, and turn slowly.

Demo 2: Shoebox Space



What you need

- A shoebox with no lid
- An **LED tealight** or small battery lantern (the Sun)
- A styrofoam or tennis ball on a stick (the Moon)
- A small paper or plush Earth on a stick
- Dark paper to line the box (optional)
- Tape and scissors. A grown-up should do the cutting!

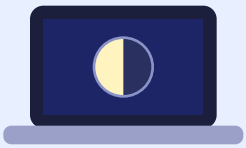
Build it and try it

1. A grown-up cuts a window hole in one short side of the box. Place the light outside, shining in through the hole.
2. Tape the **Earth** in the middle of the box floor, on a short stick.
3. Hold the **Moon** on a longer stick so it can swing around Earth in a circle. Mark 8 spots around Earth, like a clock.
4. **Full Moon:** put the Moon on the far side of Earth from the light. The side facing Earth glows!
5. **New Moon:** swing the Moon between the light and Earth. The side facing Earth is dark.
6. Take turns! Crouch so your eyes are level with Earth and look at the Moon. That is the **view from Earth**.

Bonus: swing the Moon to the top and bottom spots of its circle. You will see the half-lit **quarter moons**!

TRY IT ON A SCREEN

NASA's **Lunar Phases Interactive Lab** lets you drag the Moon around Earth and watch the phases change. Pause it at New Moon and Full Moon and compare it with your shoebox!

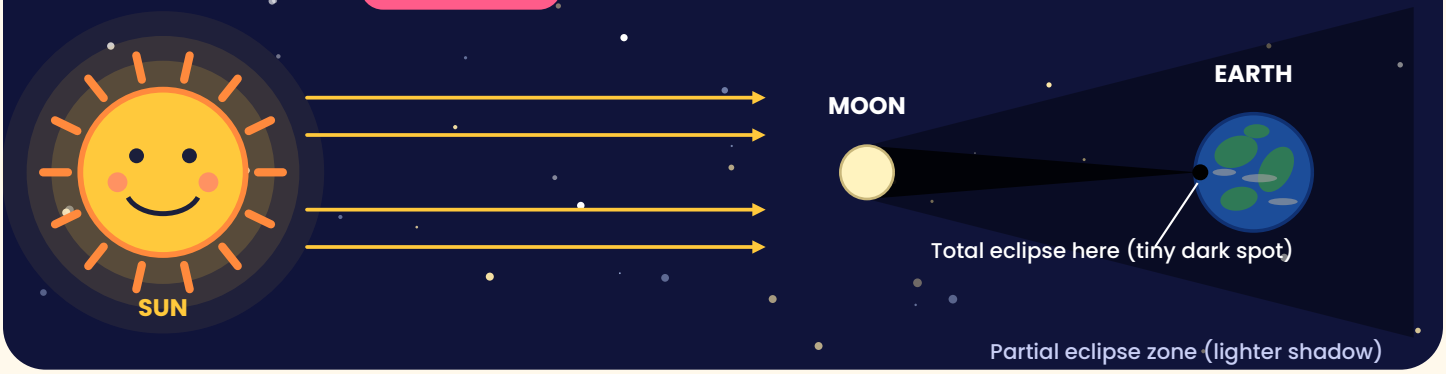


science.nasa.gov/learn/heat/resource/lunar-phases-interactive-lab/

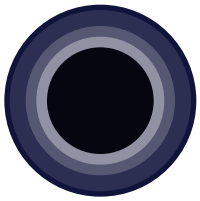
Eclipses!

SOLAR ECLIPSE

at New Moon



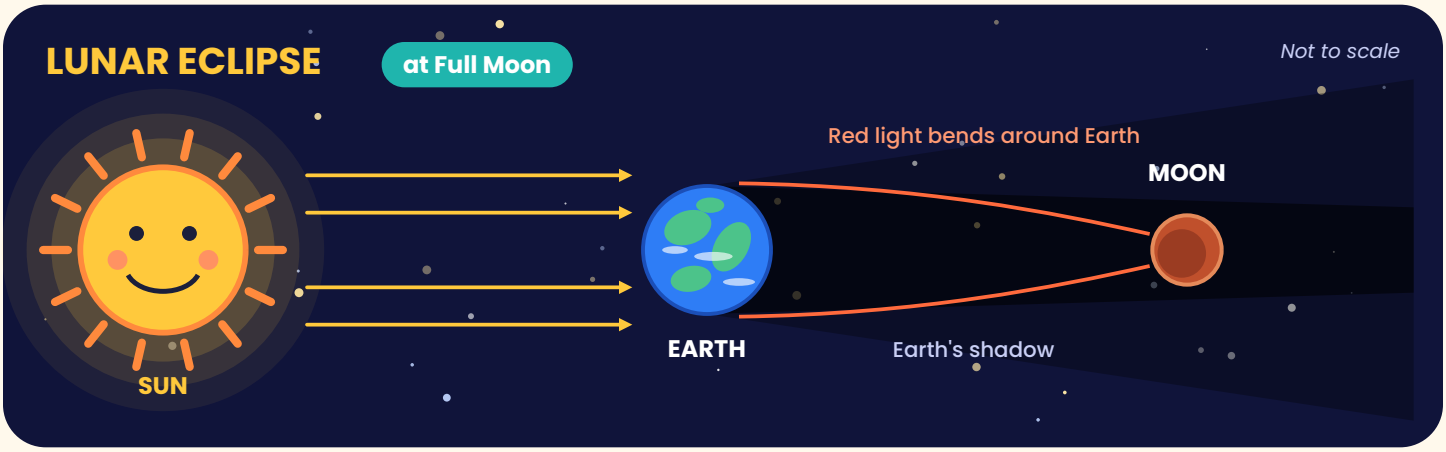
What happens: the Moon slides between the Sun and Earth, and its shadow falls on Earth.
When: only at **New Moon**.
What you see: the sky gets dim, and in the narrow dark path the Sun is completely covered, with only its glowing halo showing.



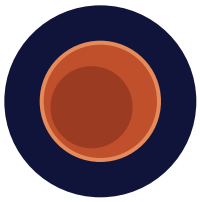
Total solar eclipse

LUNAR ECLIPSE

at Full Moon



What happens: Earth moves between the Sun and the Moon, and Earth's shadow falls on the Moon.
When: only at **Full Moon**.
What you see: the Moon dims and turns coppery red. A little sunlight bends through Earth's air and reaches the Moon, like all of Earth's sunrises and sunsets glowing at once!

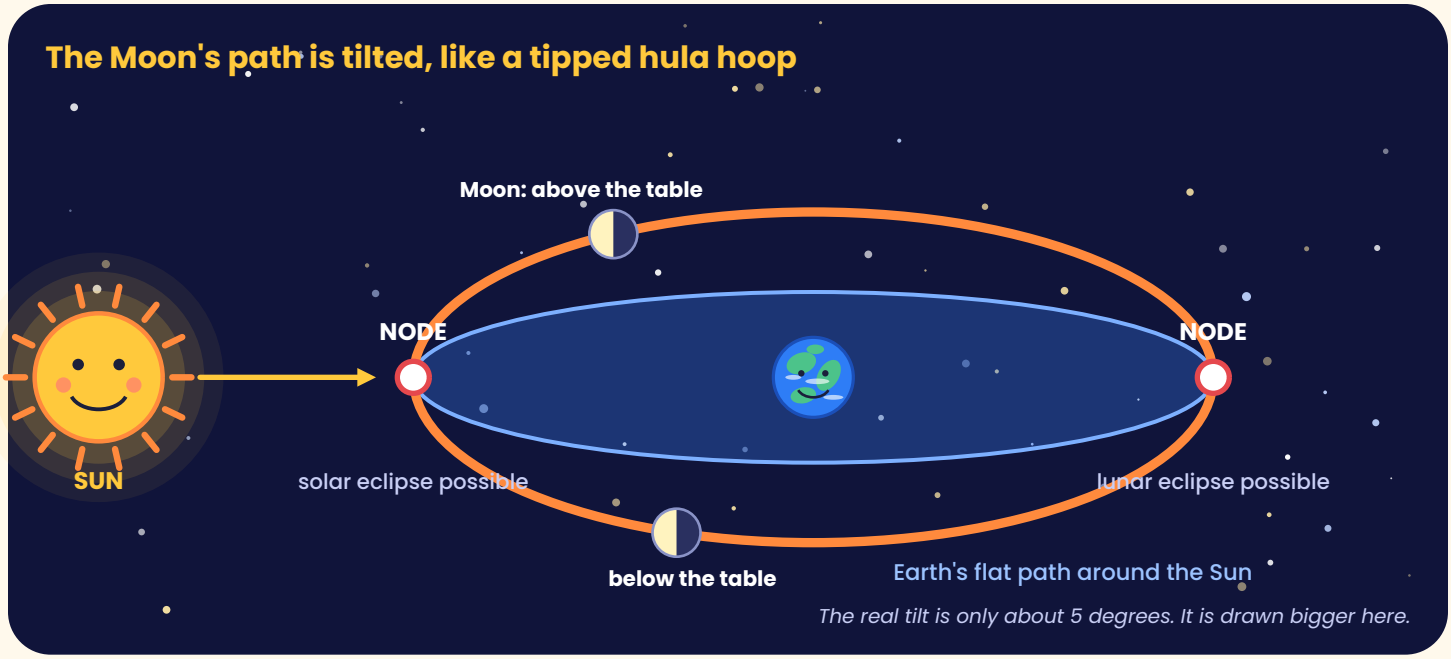


Total lunar eclipse

SAFETY FIRST

Never look straight at the Sun, even during an eclipse. Use special eclipse glasses (made to the ISO 12312-2 standard) or a pinhole projector. Regular sunglasses are **not** safe. A **lunar** eclipse is safe to watch with just your eyes.

Why Not Every Month?



Side view





So why not every month?

Because the Moon's path is tilted, at most New Moons its shadow passes **above or below** Earth. At most Full Moons, Earth's shadow passes above or below the Moon. Eclipses need the Moon to be near a **node**, where its path crosses the flat table.

Eclipse seasons

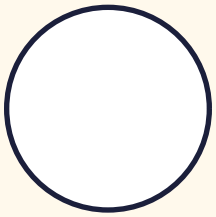
The Sun lines up with a node about **every 6 months**. For a few weeks then, a New Moon can bring a solar eclipse and a Full Moon a lunar eclipse. All around the world, there are **4 to 7 eclipses** each year (solar and lunar together).

TRY IT

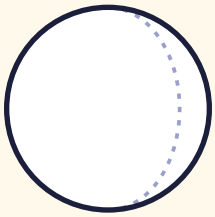
Lay one hula hoop flat on the floor. Hold a second hoop of the same size, tilted, so that the two cross at just two spots. Those two spots are the **nodes**!

Your Turn: Shade the Moon!

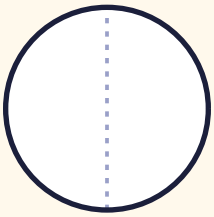
Each Moon below is a blank circle. Use a pencil to **shade the dark part** of each phase. The dotted line shows where the sunlight stops. Check your answers on page 4!



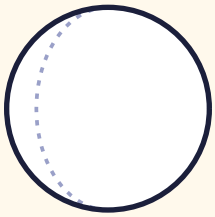
New Moon
Shade all of it!



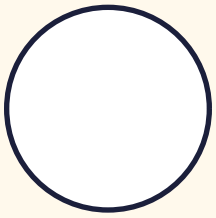
Waxing Crescent
Lit sliver on the right



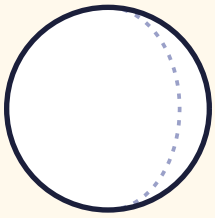
First Quarter
Lit half on the right



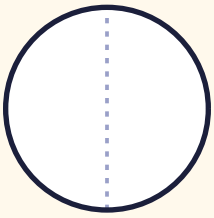
Waxing Gibbous
Lit part on the right, mostly lit



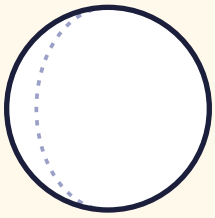
Full Moon
Leave it white!



Waning Gibbous
Lit part on the left, mostly lit



Third Quarter
Lit half on the left



Waning Crescent
Lit sliver on the left

My Moon Diary

Look at the Moon at about the same time each evening and draw what you see. Cloudy? Just write "cloudy"!

Night 1	Night 2	Night 3	Night 4	Night 5	Night 6	Night 7
Date: _____	Date: _____	Date: _____	Date: _____	Date: _____	Date: _____	Date: _____
Looks like:	Looks like:	Looks like:	Looks like:	Looks like:	Looks like:	Looks like:

After about 7 nights, you will have seen the Moon go a quarter of the way through its cycle!

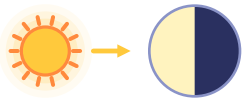
More Moon Fun

Moon Diary



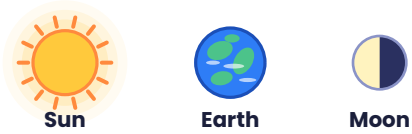
Look at the Moon at the same time each evening for a month. Draw its shape and write the date. Can you spot **waxing** and **waning**?

Shadow-Ball Art



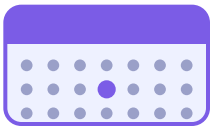
In a dark room, shine a flashlight on a ball and draw or paint what you see. Turn the ball around. Which side stays dark?

Space Hide-and-Seek



Three players: the **Sun** holds a lamp and stays still, **Earth** turns slowly in place, and the **Moon** walks around Earth. Swap roles each round!

Count the Days



Mark a New Moon on a calendar, then count the days until the next one. It takes about **29.5 days**. That is where the word "month" comes from: "moon"!

SCALE MODEL



If Earth were a **basketball**, the Moon would be a **tennis ball** about **7 meters (24 feet)** away! Space is very big and mostly empty, so our diagrams are not to scale.

NOTES FOR GROWN-UPS

- **Hemispheres:** in the Southern Hemisphere the Moon looks flipped. A waxing Moon is lit on the left there.
- **Two kinds of month:** the Moon circles Earth in about 27.3 days, but New Moon to New Moon takes about 29.5 days, because Earth also moves around the Sun while the Moon travels.
- **Common mix-up:** phases are not caused by Earth's shadow. See the myth buster on page 2.
- **Safety:** use LED lights, not candles. For a solar eclipse, use certified eclipse glasses (ISO 12312-2).
- **Going further:** after the lamp demo, ask kids to predict what the Moon will look like one week from tonight.

Moon Cheat Sheet

Why does the Moon change shape?	We see different amounts of its sunlit half as it moves around Earth. The Moon makes no light of its own.
What is a New Moon?	The Moon is between the Sun and Earth. Its lit side faces away from us, so we see a dark sky. (Sun, Moon, Earth)
What is a Full Moon?	Earth is between the Sun and Moon. The lit side faces us, so we see a bright circle. (Sun, Earth, Moon)
How long from one New Moon to the next?	About 29.5 days.
Does the Moon go around Earth in 24 hours?	No! It takes about 27 days. 24 hours is how long Earth takes to spin once. In a day, the Moon moves only about 13 degrees.
What is a solar eclipse?	The Moon blocks the Sun. It happens at New Moon. Use eclipse glasses!
What is a lunar eclipse?	Earth's shadow falls on the Moon, turning it coppery red. It happens at Full Moon and is safe to watch.
Why is there no eclipse every month?	The Moon's path is tilted about 5 degrees, so shadows usually miss. Eclipses happen near the nodes, about every 6 months.

THE ONE-SENTENCE DEMO

Put a lamp on a table (Sun), stand as Earth, and hold a ball (Moon) at arm's length. Face the lamp for a **New Moon**, then turn slowly and watch the light grow to a **Full Moon**!